Replication of: ***An Effective Method for Detecting Potential Woodland Vernal Pools Using High-Resolution LiDAR Data and Aerial Imagery (Wu et al. 2014)***

for the Town of Lunenburg, Massachusetts

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Advanced GIS ES-6100

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Abstract: Effective conservation of woodland vernal pools depends on being able to efficiently locate and map their locations with precision. Current methods for identifying potential vernal pools are primarily based on photo interpretation and digitization from aerial imagery. The success of these methods is variable in terms of accuracy and repeatability. In this paper, I follow the method proposed by Wu et al [1] --- an effective and efficient method for detecting and mapping potential vernal pools using stochastic depression analysis with additional geospatial analysis. This method was designed to take advantage of increasingly available LiDAR data that has not been frequently utilized in vernal pool studies. I successfully detected more than 300 potential vernal pools in a 72 km^2 study area in north central Massachusetts. The accuracy assessment indicated that the commission rates ranged from 14.6% to 32.7%. These rates are lower than most reported errors of previous vernal pool studies conducted in the northeastern United States. This semi-automated approach for vernal pool identification may reduce inconsistencies and difficulty associated with manual photo interpretation methods. Another strength of this analysis is the extraction of vernal pool boundaries as polygon features to analyze their geometric properties, which are not available in the current statewide vernal pool databases in Massachusetts. This study serves to validate the methods of Wu et al. and contribute to their goal of utilizing stochastic depression analysis to create a robust, comprehensive vernal pool database for the northeastern United States. A secondary goal is to provide the detected depression results to the town of Lunenburg for verification and certification so that these critical landscapes may be afforded some protection from development.

1. Introduction

Vernal pools are typically referred to as temporary bodies of water occurring in a basin depression without a permanent outflow or outlet. These depressions become filled pools in the fall, winter, or spring, and tend to dry up in the summer [1,2]. Common names for vernal pools include: ephemeral ponded wetlands, temporal ponds, or seasonal woodland ponds [1,2]. Vernal pools are often lacking breeding fish populations and are important habitats for many amphibians and reptiles adapted to these predator-lacking conditions, including many rare species [1-3]. Species assemblages in vernal pools are often more species-rich than those found in permanent waterbodies [4]. Vernal pools across the northeastern United States are at risk of destruction and degradation partly due to their short hydroperiods and quick response to changes in hydrological regimes (variability of rainfall) [1,3,4].

Several states have employed regulations regarding vernal pools and their protection, including: New Jersey, Maine, and Massachusetts. Of these states, Massachusetts was the first state in the United States to establish regulatory protection for vernal pool habitats in 1987 [2]. Massachusetts potential vernal pools undergo a certification process whereby biological data are collected by citizens and, subsequently, regulatory committees certify the pool. The collected data must include proof of use by facultative or obligate vernal pool amphibian species. These pools only become jurisdictional once approved by the state regulatory agency under the Wetlands Protection Act (WPA) [1,4].

Lack of an efficient method to identify vernal pools and other geographically isolated wetlands on the landscape undermines measures to protect these areas. Studies typically include visual interpretation of aerial photographs, image classification schemes, and field verification procedures [1,4,5]. However, these methods have limited success and commission errors (misinterpretation) vary from <3-90% in some cases. Commission errors vary depending on skill of the interpreters, imagery types, imagery acquisition dates, extent and type of land cover, and topography [1,5].

The use of high-resolution light detection and ranging (LiDAR) data can aid in the detection of vernal pools. This paper follows the methods used by Wu et al. (2014) [1] to detect potential vernal pool depressions using LiDAR data and near-infrared aerial photographs. For the purposes of this study, vernal pools were defined as a geospatially isolated depression greater than or equal to $50m^2$, with no permanent inlet or outlet that is not connected or adjacent to other surface water bodies. Stochastic depression analysis using a Monte Carlo approach was used to extract depressions from a 1-m LiDAR digital elevation model (DEM). The Monte Carlo approach estimates the likelihood of a depression occurring on the landscape by incorporating uncertainty in the LiDAR DEM (associated with data collection) [1,6]. This approach diminishes the likelihood of artifact "false" depressions being identified. The output of the stochastic depression analysis can be refined using the National Hydrography Dataset (NHD) [7], land use data, and the Normalized Difference Water Index (NDWI) [8], to identify potential vernal pools [1]. The purpose of this study is to aid in the identification of vernal pool habitat in the town of Lunenburg Massachusetts in Worcester County. By doing so, this might help the conservation commission of the town prioritize areas that can be surveyed and certified, providing this critical habitat some legal protection from development.

2. Study Area and Data

2.1. Study Area

The Natural Heritage and Endangered Species Program (NHESP) in Massachusetts has two datasets pertaining to vernal pool habitats, Certified Vernal Pools (CVPs) and Potential Vernal Pools (PVPs). These datasets don't include the boundary of the vernal pools, but rather, records are recorded as points with geographic coordinates. As of October 2018, there were 7,614 certified vernal pools in Massachusetts. The dates of vernal pools certified range from 31 May 1988 to 29 October 2018. Figure 1 shows how the number of certifications varies by year. The statewide PVP database contains 29,723 potential vernal pools [4]. The NHESP conducted the statewide PVP survey by interpreting potential vernal pools from 1:12,000-scale CIR leaf-off aerial photographs from the spring of 2000 [1,4].

It should be noted that the potential vernal pools identified in this survey do not receive protection under the WAP regulations. Evidence of amphibians or invertebrates

using a vernal pool, in addition to proof that the pool does not support a reproducing fish population, must be provided to the National Endangered Species Program (NHESP) for certification to obtain official standing as a certified vernal pool [4]. Both the CVP and PVP layers can be obtained from the Massachusetts Office of Geographic Information (MassGIS) website [9]. The data layers were prepared in shapefile format, where each point in the dataset represents the location of a vernal pool. The combined vernal pool density map for all towns of Massachusetts is shown in Figure 2 (including CVPs + PVPs).

The study area included the town of Lunenburg of Worcester County in Massachusetts (Figure 3), with a total area of 72 km^2. The town includes extensive forest/woodlands and wetlands/water bodies, with some urban/suburban development, and elevations ranging from 87 to 231m above mean sea level. As of October 2018, the study area contained 18 CVPs and 135 PVPs (Table 1). There is a fairly uneven distribution of these CVP and PVP locations throughout the study area (Figure 4). The PVPs and CVPS appear to be most prevalent on the eastern and western sides of the town with relatively few in the central region, likely due to development.

Figure 1. The number of vernal pools certified by the Natural Heritage and Endangered Species Program (NHESP) in Massachusetts.

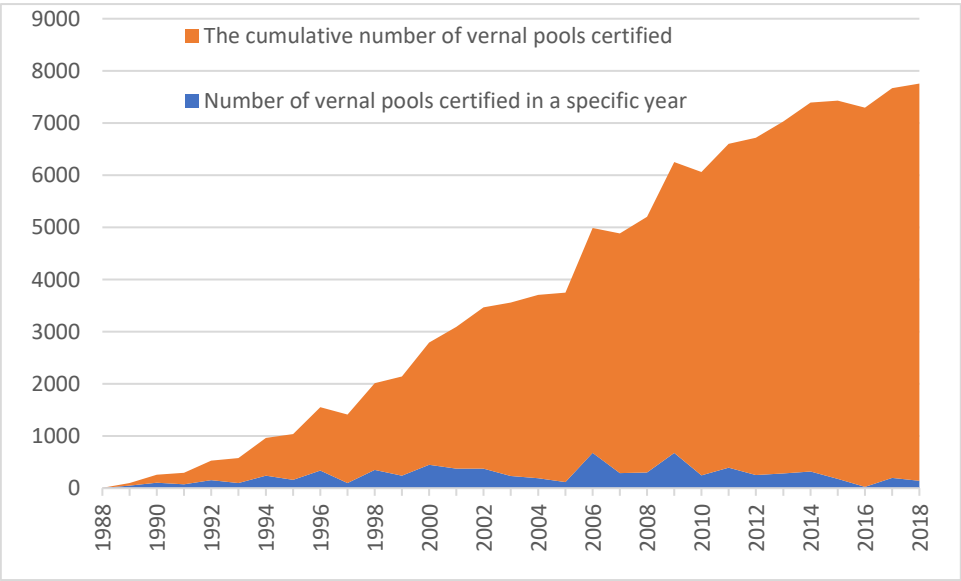


Figure 2. The combined vernal pool (CVP + PVP) density map in towns of Massachusetts as of October 29, 2018. CVPs—certified vernal pools; PVPs—potential vernal pools; NHESP—Natural Heritage and Endangered Species Program.

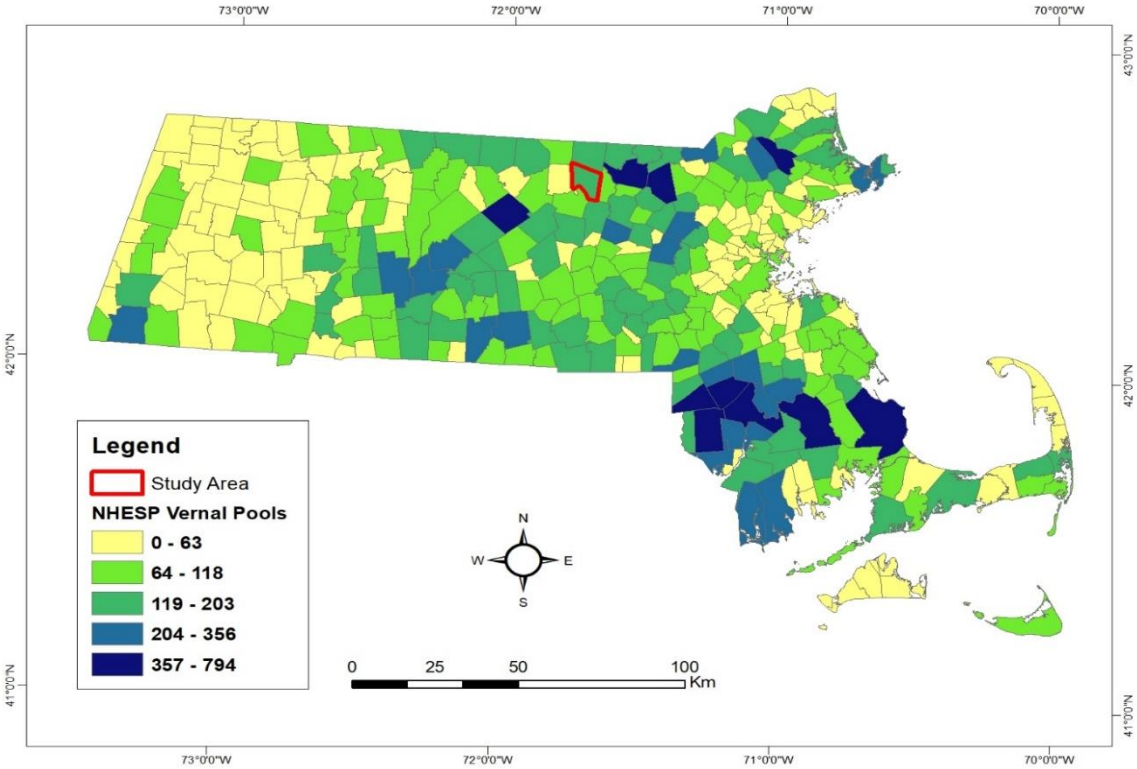


Figure 3. Location of the study area—the town of Lunenburg, in Worcester County, Massachusetts. The imagery is a false-color composite image mosaic from U.S. Geological Survey color orthoimagery acquired in April 2013.

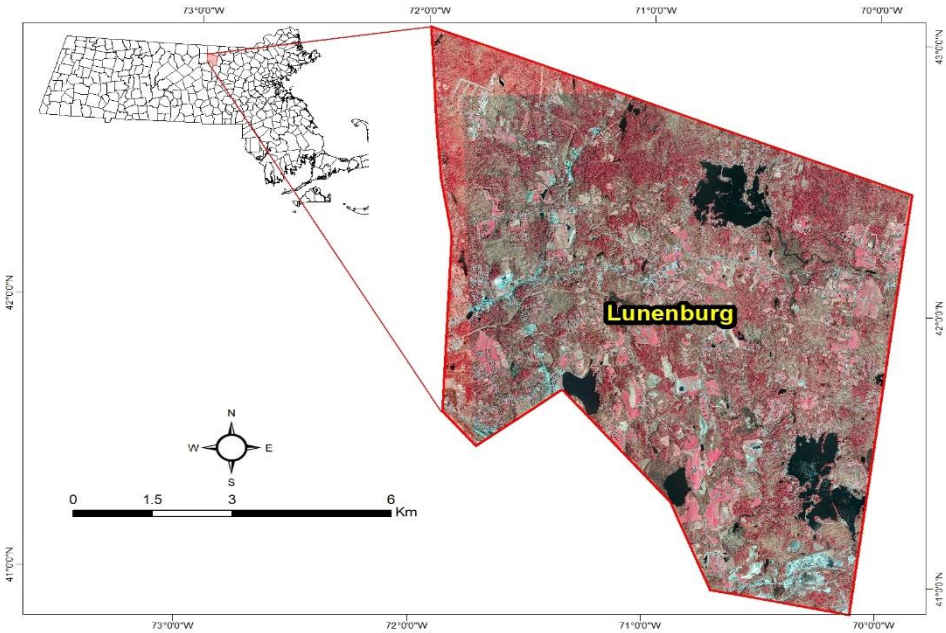


Figure 4. The distribution of Natural Heritage and Endangered Species Program (NHESP) certified/potential vernal pool locations across the study area.

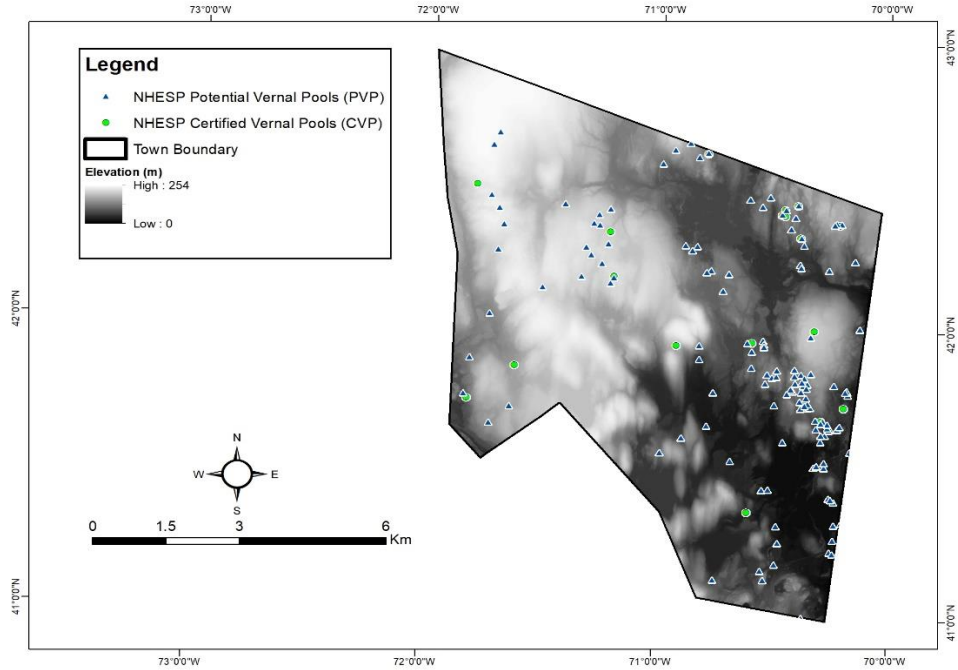


Table 1. The number of CVPs and PVPs in town of Lunenburg, Massachusetts. CVPs—certified vernal pools; PVPs—potential vernal pools.

	Lunenburg
CVPs	18
PVPs	135
Total	153

In addition to the NHESP CVP and PVP layers I used a number of ancillary layers in the analysis, including 1-m LiDAR DEM (collected in 2011), 0.3-m U.S. Geological Survey (USGS) color orthoimagery (collected in 2013), land use (developed in 2005) [10], and the NHD. All data were obtained from the MassGIS website. All these acquired datasets were projected to a common spatial reference system: North American Datum 1983 State-Plane Massachusetts Mainland FIPS 2001. A summary of the data sets used in the study is given in Table 2.

Table 2. Summary of data sets used in this study.

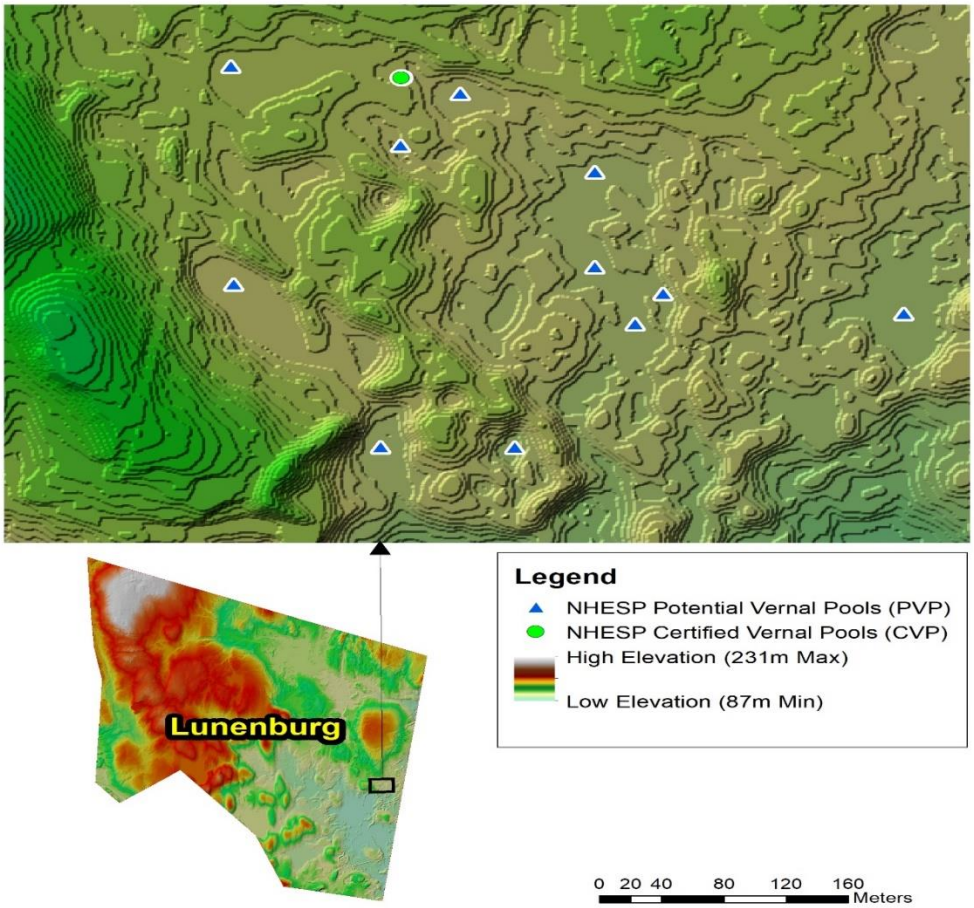
Data Set	Acquisition Date	Application
1-m LiDAR DEM	6-7 May 2011	deriving surface depressions
0.3-m color orthoimagery	April 2013	refining surface depressions
Land use data	April 2005	refining surface depressions
National Hydrography Dataset (NHD)	2005-2009	refining surface depressions
Certified vernal pool database	May 1988-October 2018	validating results
Potential vernal pool database	April 2000	validating results

2.2. 1-m LiDAR DEM

The LiDAR data were collected by the Federal Emergency Management Agency (FEMA) for its 5-year program for Risk Mapping, Assessment, and Planning (Risk MAP). The metadata for this LiDAR can be viewed through MassGIS for each fly-over area [11].

Using an Optech Gemini LiDAR system, a total 35 flight-lines of high density (Nominal pulse Spacing of 2.0m) were collected over the Nashua Watershed area. A total of 530.3 square miles was collected. A total of 3 missions were flown between May 6 and May 7, 2011, with two-meter nominal spacing that tested at 0.172m fundamental vertical accuracy at a 95% confidence level [11]. The bare-earth LiDAR DEM are available for download in GeoTiff format through Massachusetts' OLIVER as 1500m×1500m tiles with one-meter pixel resolution. The study area (Lunenburg) was composed of 50 tiles. The Mosaic to New Raster tool in ArcGIS (ESRI, Redlands, CA, version 10.6.1) was used to create a mosaicked dataset that combined the 50 DEM tiles as a seamless 1-m raster for all subsequent image analysis and map generation (Figure 5).

Figure 5. The bare-earth LiDAR DEM shaded relief of Lunenburg, Massachusetts.



2.3. 0.3-m USGS Color Orthoimagery (2013)

USGS color orthoimagery (2013) with a 0.3m resolution was acquired in mid to late April of 2013. Images were downloaded in JPEG2000 format with a 20:1 compression ratio. Images contain a four-band (red, green, blue, and near-infrared) spectral classification, and 1500m×1500m extent. The study area was composed of 50 tiles. The Mosaic to New Raster tool in ESRI ArcGIS 10.6.1 was used to create a mosaicked dataset that combined the 50 image tiles as a seamless four-band, 0.3-m image, which was used for all subsequent image analysis and map generation (Figure 3). The Color-Infrared (CIR) orthoimagery was used both for visual assessment/verification of vernal pool locations and to create NDWI and NDVI images from the spectral bands to identify water features associated with pools [1].

2.4. Land Use Data (2005)

MassGIS provides a statewide digital vector dataset of land use categories based on 0.5-m resolution four-band orthoimagery acquired in 2005 [10]. The original MassGIS land use data layer has 40 land use categories, which were aggregated into five general classes for the study, including forest (including successional), developed, wetlands, grassland (agriculture or pasture), and water (Table 3).

Table 3. The land use composition of the study area.

Class	Area (km^2)	Percentage
Forest	40.91	56.81
Developed	16.24	22.55
Grassland	6.12	8.50
Wetland	4.98	6.92
Water	3.75	5.21

2.5. National Hydrography Dataset (NHD)

Following Wu at al. [1] and Lane et al. [12], buffer zones were created from the NHD data to separate potential "geographically isolated" depressions from depressions assumed to be connected to streams or permanent waterbodies, based on proximity or adjacency. Vernal pools and ephemeral wetlands are often hydrologically, biologically, and/or biogeochemically connected to other waters, however, this varies on a case-by-case basis [1,12]. For the purpose of this analysis, all potential depressions immediately adjacent to or within streams and other surficial waterbodies were removed. Using the 1:24,000-scale NHD data as the source data for hydrology, I created a 10-m distance buffer from the NHD flowlines. Two Select by Locations were performed within ArcGIS 10.6.1 to remove depressions connected to waterbodies. Depressions within the NHD waterbodies polygon (lakes and ponds) were excluded from further analysis. A second Select by Location was performed to remove depressions located within and intersecting the 10m buffer from stream flowlines.

3. Methods

3.1 Stochastic Depression Analysis

For this study, vernal pools are defined as confined surface depressions with no permanent surface inflow or outflow and a total area equal to or larger than $50m^2$. Per Wu et al. [1], potential vernal pool identification methods involved extracting surface depressions from the bare-earth LiDAR DEM, followed by further refining the

depressions by eliminating those that occurred on developed lands or were identified as permanent waterbodies (NHD layers). Surface depressions located in forest, wetland, or grassland land-use designation with identifiable water features would be designated as potential vernal pools for subsequent analyses.

The Stochastic Depression Analysis Tool in Whitebox Geospatial Analysis Tools (GAT) [13] was used to derive depressions from the LiDAR DEM. Whitebox GAT is an open-source GIS and Remote Sensing software package developed at the University of Guelph's Centre for Hydrogeomatics. The program (GAT) and can be downloaded from the Whitebox website [1,13]. The Stochastic Depression Analysis Tool can be used to map topographic depressions in a DEM, taking into account the uncertainty in depression shape resulting from DEM error [13]. The tool uses a Monte Carlo approach to map depressions. Based on the Monte Carlo procedure, each grid cell in a DEM is assumed to follow a Gaussian error probability distribution function (PDF) with a mean of zero and a specified standard deviation (error magnitude) [14]. The LiDAR data root mean squared error (RMSE) value can be used to represent the error magnitude of the data. The RMSE value of the LiDAR in the study area is 0.088 meters, which was specified in the metadata [11]. I used the dnorm function in R Statistical Software (mean=0, standard deviation=0.088m) to create a distribution of the probability of LiDAR data error (Figure 6). Once the distribution of the probability of LiDAR data error was created, a random sample was drawn from the Gaussian error PDF of each grid cell and added to the original DEM. Depressions in the error-added DEM were then filled using the highly efficient depression filling algorithm developed by Wang and Liu [15]. In the Stochastic Depression Analysis Tool dialogue box this is known as "histogram matching". A text file containing two columns (RMSE value and frequency) generated from the frequency distribution of the error are input into the tool. Note that Figure 6 shows the probability of an error value, not the frequency as input into the tool.

The user can specify the number of iterations in which the depressions are mapped and with each iteration, a new random draw of error values are mapped to each cell. The probability of a grid cell belonging to a depression feature, given the uncertainty (RMSE), was mapped using an 80% probability as the threshold. For instance, for a depression to be delineated as a "true depression", it must be identified as a depression for at least 4 out of 5 iterations. Following Wu et al., I used 50 iterations due to the length of computation run-time (roughly 1 hour per iteration).

Other required fields for the Stochastic Depression Analysis Tool include number of turning bands and the range to be used in the simulations. Turning bands refer to a particular type of simulation to create error fields (the error associated with the RMSE). Also known as the random lines technique or turning bands simulation, it employs a number of one-dimensional simulations to develop a three-dimensional final output. Each one-dimensional linear simulation is treated as a three dimensional vector that passes through the center of a theoretical sphere, with the orientation represented as (x,y,z), where each variable is assigned random numbers. This method of generating error fields

in the DEM is computationally inexpensive compared to other methods [16]. A thousand turning bands were input into the tool for this study.

The range is extracted from the input DEM's error semivariogram (Figure 7) and determines the spatial correlation length (the distance over which individual error values within the grid are correlated). The semivariogram was generated in ArcGIS using Geostatistical Analyst. Semivariograms can only be generated from vector data, therefore, 1,000 random points within the study area were generated using the Create Random Points Tool. Then, the Extract Values to Points Tool provided the elevations of the grid cell in the DEM containing those points. From there, the semivariogram was generated using the elevation values assigned to the random points. A model or regression was fitted to the semivariogram to calculate the approximate range value. Simple Kriging in the Geostatistical Analyst Wizard allows a regression line to be fit through the semivariogram to determine the correlation between points (Figure 8). The range of the DEM values was determined to be approximately 4,443m. The regression line also validated the 8.8cm RMSE value.

Following the detection of the depressions, the depression raster was reclassified into a binary grid where depression pixels were assigned a value of 1 and all other pixels a value of 0. The Fill Tool removed erroneous pixels and imperfections (holes) in the depressions. The Raster Cleanup menu of ArcScan and the Close Tool (morphology operator) smoothed the boundaries of the depressions by Eroding and Dilating the regions (adding and removing pixels from the boundaries). The Region Group tool in ArcGIS was then used to group depression pixels using their 4 immediate neighbor cells. The resulting depression pixel objects were labeled and indexed with a unique integer number. These uniquely identified raster objects were then converted to vector format as polygons and the boundaries of the depression features were extracted for subsequent analysis.

Figure 6. Probability distribution of LiDAR data error.

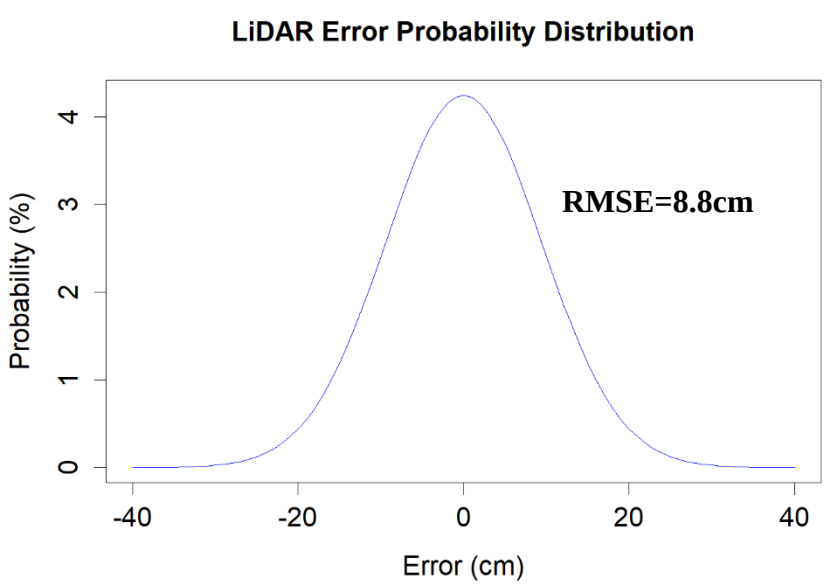


Figure 7. Semivariogram of 1,000 random points selected from LiDAR DEM.

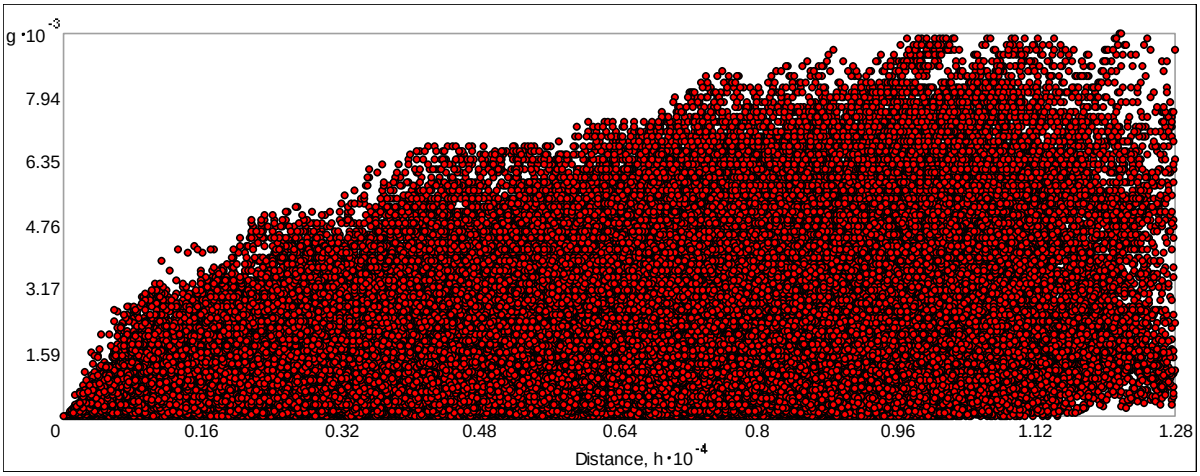
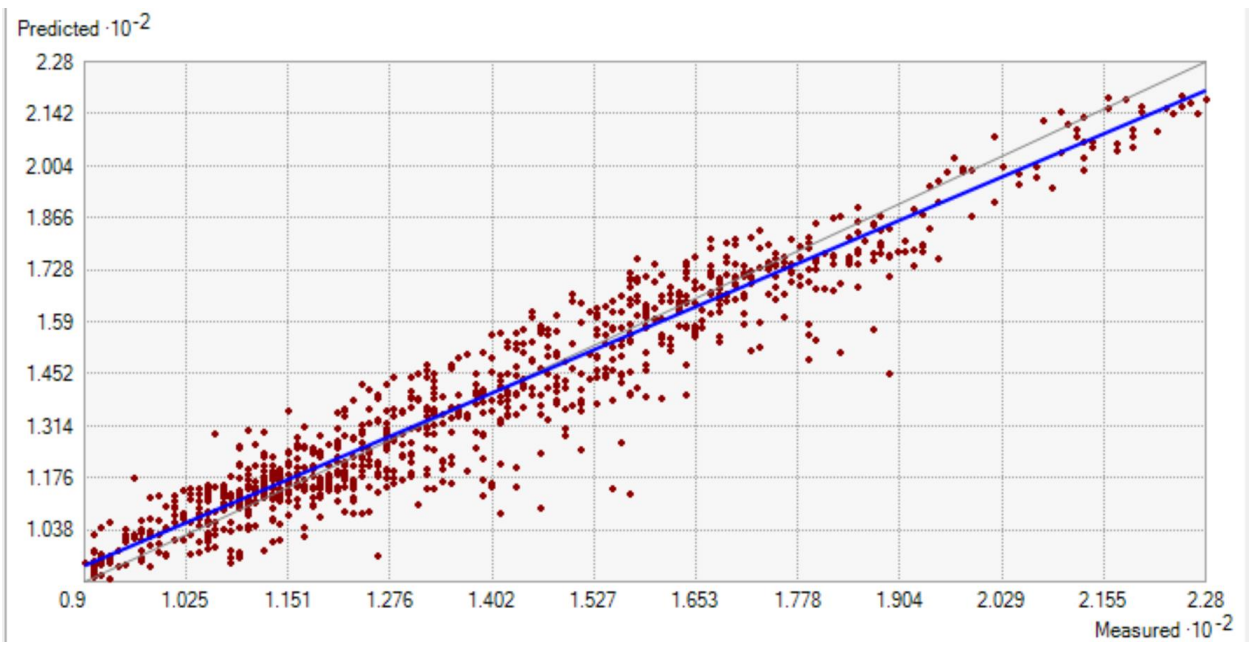


Figure 8. Regression Model of semivariogram values using Simple Kriging. Model used to calculate Range for Stochastic Depression Analysis and validate RMSE value.



3.2. Eliminate Depressions Occurring in Permanent Waterbodies and Developed Land

Using the definition of "geographically isolated wetland with no inflow or outlet", vernal pools should not be connected to permanent hydrologic features, such as rivers, streams, lakes, and reservoirs. Select by Location was used to find depressions within permanent waterbodies such as lakes and ponds and these depressions were removed. Then, a second Select by Location on the 10-m buffered NHD water polyline layer

(streams) was used to select depression polygons adjacent to streams. Depressions not overlapping any NHD water features were retained for further analysis, while the others were considered adjacent or connected waters and removed from further analysis or consideration as possible vernal pools [1,12].

I intersected the remaining depressions with the reclassified land use data layer (developed, forest, grassland, wetlands, and water) and calculated the land use modality [1] for each depression using Zonal Statistics as Table. The maximum percentage land use within each depression was designated as the dominant land use type. Only those depressions with forest, grassland or wetlands as their dominant (50% or greater) land use types were retained for further analysis.

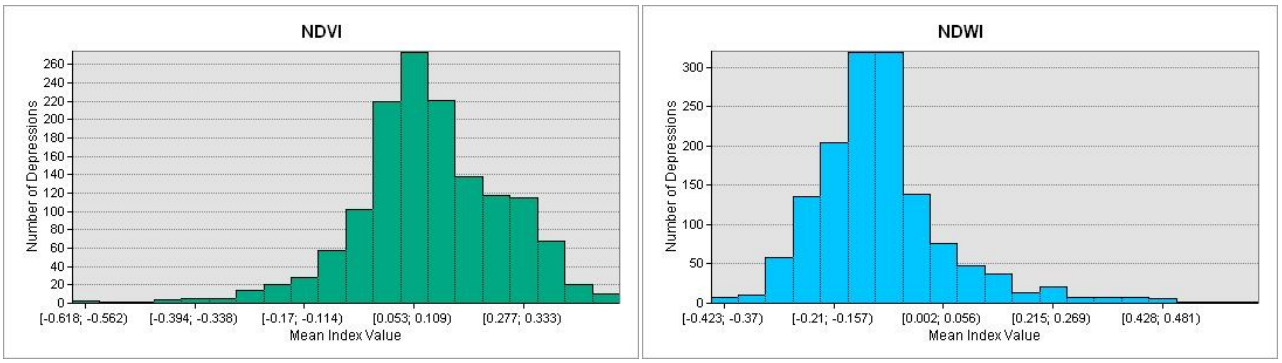
3.3. Refine Depressions Using the Normalized Difference Water Index (NDWI)

The NDWI identifies open water features in wetland environments with remotely sensed imagery. The NDWI uses reflected near-infrared radiation and visible green light to enhance the presence of water features while eliminating the appearance/presence of soil and terrestrial vegetation features [1, 17]. The NDWI is calculated as follows and this equation can be performed using Raster Calculator in ArcGIS:

$$NDWI = \frac{Green - NIR}{Green + NIR}$$

In the output NDWI raster, the water features will have positive values while soil and vegetation have zero or negative values. In this study, the CIR orthoimagery was used to calculate both the NDVI and NDWI images. The Zonal Statistics tool in ArcGIS was used to derive the maximum NDWI (NDWI MAX), averaged NDWI (NDWI AVG), and averaged NDVI (NDVI AVG) of the depression pixels. Values for each of the refined 1,420 depressions from the depression polygon layer for NDWI and NDVI AVG are shown in Figure 9. The distribution patterns of the NDVI AVG and NDWI AVG were relatively symmetric around the mean. Depressions with high NDVI AVG tend to have low NDWI AVG, and vice versa. NDWI MAX was used along with NDWI AVG to validate that water pixels existed inside depressions. The use of NDWI MAX can extract those depressions with water pixels, while the use of NDWI AVG can ensure sufficient abundance of water pixels inside depressions, which can eliminate depressions with only a few water pixels. Consistent with Wu et al., I chose NDWI MAX > 0.3 and NDWI AVG > −0.15 as the selection criteria for these values. I performed a Select by Attribute to select depressions that met these criteria. These depressions were designated as potential vernal pools.

Figure 9. The histograms of averaged NDVI and NDWI for 1,420 depressions.



4. Results

4.1. Validation/Classification with the NHESP, CVP, and PVP Databases

I compared the depression analysis results to the NHESP statewide vernal pool databases (PVP and CVP). The potential vernal pools were not field-verified [4], but the PVP database can still serve to validate this study's methods for vernal pool identification [1]. Validation was performed by overlapping vernal pools with the depression detections by overlaying the statewide vernal pool databases and the mapped vernal pools in ArcGIS onto the USGS color orthoimagery. There are differences in the precision with which the locations of vernal pools were mapped in both PVP and CVP data layers. PVPs have a precision of ≤ 5m from the center of an interpreted pool, while the locations of certified vernal pools have a precision of approximately 15m in any direction [4]. Therefore, PVPs and CVPs were handled separately for detection analysis.

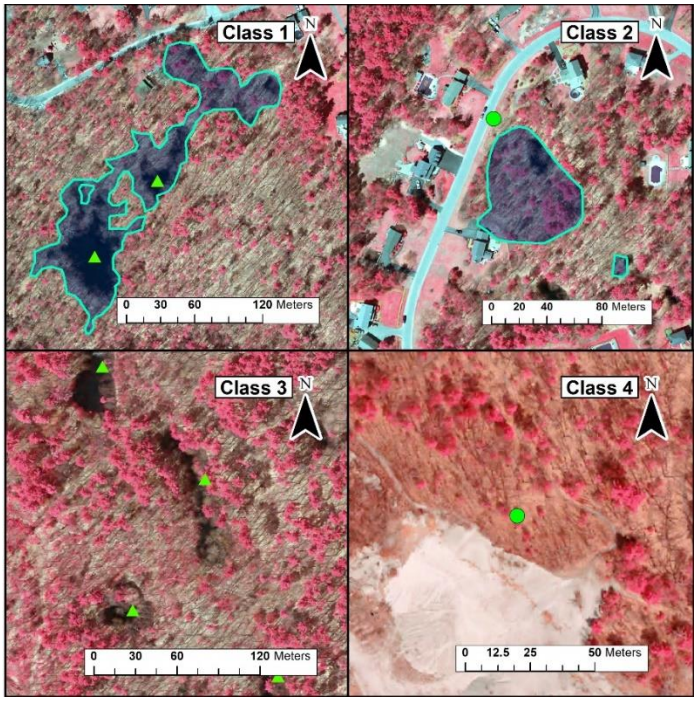
CVP were buffered 15m, while PVP were buffered 5m. CVP and PVP spatial relationships to the detected depressions were classified into four categories (Table 4): Class 1, present vernal pools that were directly located inside detected depressions; Class 2, present vernal pools that were not contained by any depressions, but were within the 15m/5m buffer distance; Class 3, present vernal pools that were not detected; Class 4, vernal pools that were absent or not identifiable from the CIR orthoimagery (missing water). Class determination of the CVPs and PVPs was conducted using photo interpretation and Select by Location in ArcMap. CVP and PVP were only considered vernal pools if they were associated with a water feature on orthoimagery. If CVPs/PVPs were not contained within a depression, a buffered planar distance of 15m/5m around the point was analyzed to determine proximity to the depression. If there was more than one depression within the 15m/5m buffer distance of the point, the closest one was chosen to be correlated with the vernal pool [1]. Vernal pools were considered absent if an identifiable water feature could not be associated with the feature on the map. These could be areas of development or where vernal pools have dried up and no longer exist.

Of the 153 statewide CVPs and PVPs in the study area, 33 (21.6%) were located inside detected depressions, and 7 (4.6%) were located within 15m/5m of depressions. My methods did not detect 102 (66.7%) of pools visually confirmed to be present, while 11 (7.2%) CVP + PVP vernal pools were considered as absent (or "missing"), as no identifiable water features could be observed from the orthoimagery. The combination of Class 1 and Class 2 indicate that there was an overlap of 40 (26.2%) pools between detected depressions and the statewide vernal pool databases.

Table 4. The classification of certified vernal pools (CVPs) and potential vernal pools (PVPs).

Class ID	Class Description	Count	Percentage	Cumulative Percentage
1	Present CVPs/PVPs located inside depressions	33	21.6	21.6
2	Present CVPs/PVPs within buffer distance (15m/5m) of depressions	7	4.6	26.2
3	Present CVPs/PVPs not related to any depressions	102	66.7	92.9
4	Absent vernal pools	11	7.2	100

Figure 10. Examples of different certified vernal pools (CVPs) and potential vernal pools (PVPs) overlaid on color-infrared aerial photographs (2013). The green circular dots represent CVPs and the green triangles represent PVPs. The polygons with blue outlines represent detected depressions.



4.2. Depression Detection Result Analysis

Initially, 1,802 depressions with area >50m^2 were detected in the first step with Stochastic Depression Analysis. After eliminating depressions occurring in permanent waterbodies there were 1,654 depressions remaining. Removing depressions on developed lands resulted in 1,420 remaining. Further restricting the data using the NDWI MAX and NDWI AVG criteria resulted in a final 481 depressions selected as potential vernal pools in the study area.

Each depression polygon (predicted vernal pool) was examined and overlaid with layers, such as CVP, PVP, a LiDAR shaded relief map, and CIR orthoimagery, to determine detection success. By overlaying these layers, and through visual interpretation, the 481 depressions were classified into nine categories according to spatial relationships with the CVP and PVP points in Table 5 [1]. Depression polygons (Class 1–5) overlapping or proximal to CVPs or PVPs were considered as successful detections and accounted for approximately 6.8% of the depressions in the dataset. Note that there was one CVP that was located within a depression, however, this was the only pool that contained a PVP and CVP and therefore, this depression was only counted as Class 3. The main goal of this study was to identify new pools in Lunenburg including those not identified within the statewide inventory. I identified an additional 291 depressions (Class 6) that were considered successfully identified vernal pools. This suggests, photo interpretation methods alone may not be effective at detecting potential vernal pools [5].

Depressions of Class 7 (n = 75) were considered as uncertain vernal pools, as the structure of the pool could not be determined. Twelve depressions (Class 8) were deemed to be too large to be vernal pools and were considered as errors. Although the boundary of a vernal pool was not detected for Classes 7 and 8, these classes had an identifiable water feature and were a topographic depression on the landscape. Polygons in Classes 7 and 8 were considered as “partially successful” detections, as they could be examples of orthoimagery interpretation error (Class 7) or large vernal pool complexes [1,18] within a topographic depression (Class 8). Assuming Classes 7 and 8 were correctly identified, then 87.4% of the depressions correctly identified vernal pools in the landscape. Assuming that Class 7 and Class 8 polygons were incorrect, detection accuracy is reduced to about 67.3%. I was able to observe that 14.6% (n = 70) of the depressions identified were not vernal pools (Class 9). The majority of errors associated with false depression detection were associated with shadows that were confused for water (see Class 9 in Figure 11). The descriptive statistics of the 481 detected depressions (Classes 1 to 9) are shown in Table 6, including area, perimeter, mean elevation, mean NDVI, and mean NDWI. The mean size of these classes was 1,993m^2 and the mean perimeter was 183m.

4.3. Accuracy Assessment (Commission and Omission Errors)

The accuracy assessment of potential vernal pool detections was evaluated by errors of commission and omission. Errors of commission were depressions that were incorrectly identified as vernal pools. Errors of omission were vernal pools that were not accounted for. These existing vernal pools could not be detected with surface depression analysis (CVP and PVP). The commission rate was 14.6% when Classes 7 and 8 were considered as successful detections and 32.7% when they were considered to be incorrect. The number of the present CVPs and PVPs that were not identified by these methods can be used as a proxy for omission. Omission was defined as the ratio between the omitted vernal pools and the total number of vernal pools (detected vernal pools + omitted vernal pools) [1]. There were 481 detected depressions and 102 omitted vernal pools, and the proxy omission rate was calculated to be 17.5%.

Figure 10. Examples of different depression classes, as described in Table 5, overlaid on color-infrared aerial photographs (2013). The green circular dots represent certified vernal pools (CVPs), and the green triangles represent potential vernal pools (PVPs). The polygons with blue outlines represent detected depressions.

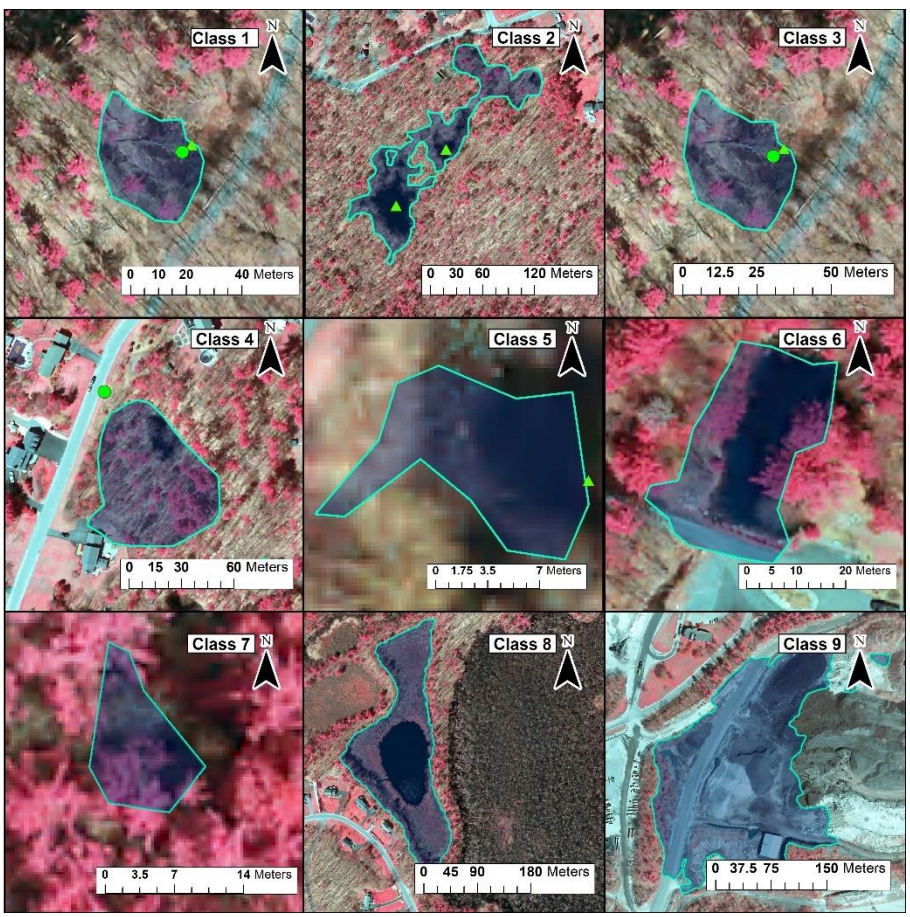


Table 5. Depression types and their relationships to statewide vernal pool databases. CVPs—certified vernal pools; PVPs—potential vernal pools.

Class ID	Class Description	Count	Percentage	Cumulative Percentage
1	Depressions that contain CVPs	0	0%	0%
2	Depressions that contain PVPs	27	5.6%	5.6%
3	Depressions that contain both CVPs and PVPs	1	0.2%	5.8%

Table 5. *Cont.*

Class ID	Class Description	Count	Percentage	Cumulative Percentage
4	Depressions within 15m buffer distance of CVPs	1	0.2%	6.0%
5	Depressions within 5m buffer distance of PVPs	4	0.8%	6.8%
6	Depression considered as certain vernal pool but not related to a CVP/PVP	291	60.5%	67.3%
7	Depressions considered as uncertain vernal pools and not related to a CVP/PVP	75	15.6%	84.9%
8	Depressions whose boundaries are bigger than vernal pools	12	2.5%	87.4%
9	Depressions not considered as vernal pools	70	14.6%	100%

Table 6. Descriptive statistics of 481 detected depressions (Classes 1–9). ELEV AVG—averaged elevation; NDVI AVG—averaged normalized difference vegetation index; NDWI AVG—averaged normalized difference water index.

Index	Mean	Minimum	Maximum	Sum
Area (m^2)	1,993	50	104,638	958,644
Perimeter (m)	183	27	4,241	87,919
ELEV AVG (m)	126.5	89.6	226.7	N/A
NDVI AVG	0.002	-0.62	0.28	N/A
NDWI AVG	-0.13	-0.37	0.40	N/A

5. Discussion

Vernal pools are important ecosystems that contribute to amphibian metapopulations, the maintenance of threatened, endangered, and rare species, water storage, and local biogeochemical cycles [1,19,20]. Vernal pools and other wetlands are typically small and shallow systems that can be overlooked and are frequently unprotected by local, state, and federal regulations [1,21]. Being able to locate these systems is the first step towards better understanding of their functions, connections to other ecosystems, importance in the landscape, and developing methods to protect them from degradation. In the past, aerial photography was the best way to identify vernal pools, but with the advent of LiDAR data (that is now readily available), and improved geostatistical tests, we have a better chance of detecting these features on the landscape. The approach by Wu et al. [1] replicated in this study was successful in identifying 291 additional vernal pools in the study area when compared with the certified and potential vernal pool databases (shown in Table 5).

Wu et al. [1] mention that discrepancies in the number of depressions identified as putative vernal pools varies according to minimum mapping units. That is to say, there is no agreement in the literature as to what area (in m^2) is acceptable as the threshold for vernal pool identification. As pointed out by Burne [4] in the statewide potential vernal pool database, only pools of 38.1 meters in diameter and larger on the photographs could be reliably identified when photos were of fair to excellent quality [1]. However, according to Wu et al. [1], stochastic depression analysis is efficient at identifying vernal pools that are smaller than studies that employed photo interpretation methods. This also indicates that photographic interpretation may underestimate the number of vernal pools.

The 70 (14.6%) false positive identifications of vernal pools in Class 9 most likely resulted from confusion between tree shadows and water bodies, due to their similar spectral signatures derived from the CIR aerial photographs [1]. This would result in positive NDWI values in the depression that caused it to be falsely identified as a vernal pool. In addition, date of the data acquisition between LiDAR, CIR aerial photographs, and land use data may have also contributed to the false identification of vernal pools

(temporal discrepancies). For the purposes of this analysis, I assumed that depressional vernal pools do not occur on developed land, and the land use data in 2005 was used to refine the depressions. The land use data was from 2005 so any recent development could lead to the removal of some artifact depressions, further refining the data. The CIR aerial photographs used in this study were acquired in generally leaf-off (lack of foliage) conditions during mid to late April 2013, but it should be noted that not all potential vernal pools were holding water during the imagery acquisition time. Some pools might already have dried out in the early spring or were just beginning to form in the early summer.

Approximately 22% of the 153 CVPs + PVPs within the study area occurred within detected depressions. However, 66.7% of the CVPs + PVPs were not associated with depressions (see Table 4) and were considered as errors of omission in the study. By inspecting the omitted pools and their associated DEM, NHD, and depression layers, these omissions could be separated into two categories. The first category was depressions associated with omitted pools that were not detected with my stochastic depression analysis in the first step (Section 3.1). These could be considered true errors of omission and may have been due to the RMSE error for the LiDAR data (0.088m) being greater than the average depth of the vernal pool system [1]. The second type are omissions which come from the refinement process in which these omitted pools were initially detected using the stochastic depression analysis, then were eliminated because as they did not meet the refinement criteria (see sections 3.2 and 3.3). The adjacency criteria of >10m from NHD flowlines seemed to account for most of these errors. Through the refinement process, 10 vernal pools were removed that were within 10m buffer zones of NHD features, which could have accounted for up to 9.8% of the errors of omission.

6. Conclusions

Wu et al. [1] developed an efficient and effective method for detecting and mapping potential vernal pools using stochastic depression analysis with additional geospatial analysis and this was successfully replicated in this study. High-resolution LiDAR data, is increasingly available and useful in identifying topographic depressions and can be employed in vernal pool studies. I successfully detected more than 300 putative vernal pools in Lunenburg, Massachusetts. The accuracy assessment indicated that commission rates might range from 14.6% to 32.7%. One significant advantage of this semi-automated approach for vernal pool identification is that it may reduce inconsistencies and alleviate repeatability concerns associated with manual photointerpretation methods [1]. Another useful aspect of this method is that the boundaries of vernal pools can be extracted as polygon features to characterize their geometry, which are not available in the current statewide vernal pool databases in Massachusetts (they are simply points). Incorporating LiDAR and depression analysis with a GIS-based approach might prove effective in discerning vernal pool ecosystems in order to inform management and conservation of these ecosystems. In conclusion, my results corroborated the results of Wu et al. [1] and demonstrate the potential of using LiDAR data along with CIR aerial

photographs for vernal pool mapping and inventory. These methods could serve to create a regional potential vernal pool database that would be valuable in enhancing our understanding of the current status of vernal pools in the northeastern United States in response to anthropogenic degradation. To follow up on this study, a standalone tool will be created using Python scripting that will allow a user to conduct this analysis by inputting parameters into a dialogue box within ArcMap. The stochastic depression analysis and refinement criteria will be calculated automatically.

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